

**BACHELOR OF COMPUTER SCIENCE (HONS)****MAY-AUG 2026 TEST****COURSE: DATABASE FUNDAMENTALS****COURSE CODE: BCC1223****TIME ALLOWED: 2 HOURS**

Total Marks: 100

Duration: 2 Hours

Part A: Conceptual Questions – 20 Marks

Answer the questions below.

1. Explain the concept of data integrity and discuss **THREE (3)** types of integrity constraints.
(4 marks)
2. Define primary key, foreign key, and composite key with suitable examples.
(3 marks)
3. Draw a simple ERD of a student registration system with minimum requirements - 3 entities, 2 attributes each, PK, FK, and cardinality appropriately. You can use either Chen's or Crow's Foot notation.
(13 marks)

Part B: DDL (Data Definition Language) – 35 Marks

Answer all questions. Use SQL statements where appropriate.

1. Write SQL commands to create a table Student with the following fields:
 - StudentID (INT, Primary Key)
 - Name (VARCHAR(50))
 - Gender (CHAR(1))
 - DateOfBirth (DATE)
 - CourseID (INT, Foreign Key referencing Course table)
(5 marks)

2. Modify the Student table to add a new column Email (VARCHAR(100)).
(5 marks)
3. Create a table Course with the following fields and constraints:
 - CourseID (INT, Primary Key)
 - CourseName (VARCHAR(100))
 - CreditHour (INT, must be between 1 and 6)(5 marks)
4. Alter the Course table to add a constraint that ensures CreditHour cannot be null.
(5 marks)
5. Drop both StudentInfo and Course tables from the database.
(5 marks)
6. Write SQL to create a Lecturer table with a unique Email and a foreign key CourseID referencing Course.
(5 marks)
7. Write SQL to create a table, Enrolment, with a composite primary key (StudentID, CourseID).
(5 marks)

Part C: SQL Query (DML and SELECT Queries) – 45 Marks

Answer all questions based on the schema below:

Student(StudentID, Name, Gender, DateOfBirth, CourseID)

Course(CourseID, CourseName, CreditHour)

Enrollment(StudentID, CourseID, Grade)

1. Retrieve the names of all female students enrolled in any course.
2. Display all students' names and their courses.
3. Retrieve the number of students enrolled in each course (display CourseName and TotalStudents).
4. Write a query to list the students who obtained a Grade = 'A' in any course.
5. Display the average CreditHour of all courses.
6. Write a query to show the youngest student's name and date of birth.
7. Retrieve a list of courses with more than FIVE (5) enrolled students.
8. Write a query to update a student's grade in the Enrollment table to 'B+' where StudentID = 101.
9. Delete all enrollment records for a course with CourseID = 500.
(5 marks each question = 45 marks)